How the Cognitive Score app works

Every step from the electrodes to the numbers on screen: **how** it is done, **why** it is done that way, and **where** each method and number comes from. Worked examples use your real recordings of 30 September 2026.

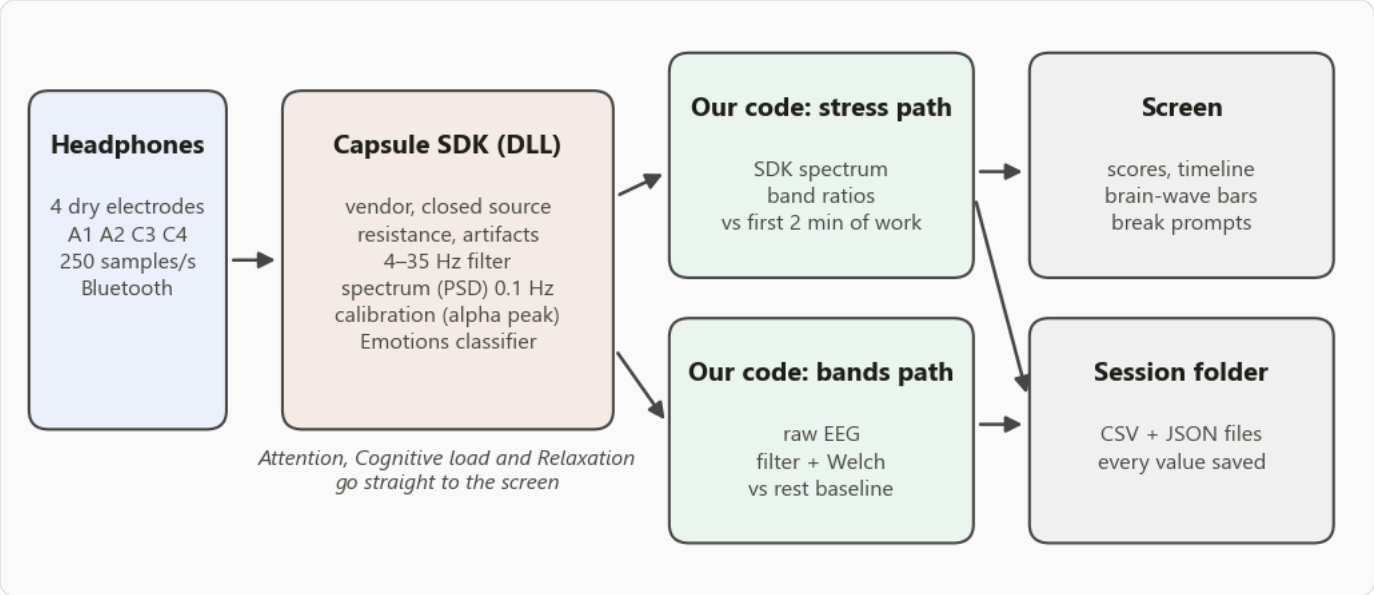
> **Reading the "From where" tags**
> Vendor the headset maker's SDK or documentation Literature a published study or standard Measured measured on your recordings or the simulator Design choice a choice made in this project, not yet validated.

1. The big picture
2. The headset and its signal
3. Connecting
4. Electrode contact
5. Calibration
6. Rest baseline
7. The SDK spectrum
8. Attention, cognitive load, relaxation
9. Stress
10. Fatigue
11. Brain-wave bars
12. Breaks and check-ins
13. Recording
14. Testing
15. Known issues
16. All numbers at a glance
17. Check yourself
18. Sources

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1. The big picture

The app is a chain of small steps. The headset measures voltages; the vendor's software (the Capsule SDK, a closed-source Windows DLL) cleans them and computes some scores; our Python code computes the rest, decides when to prompt you, draws the screen and saves everything.

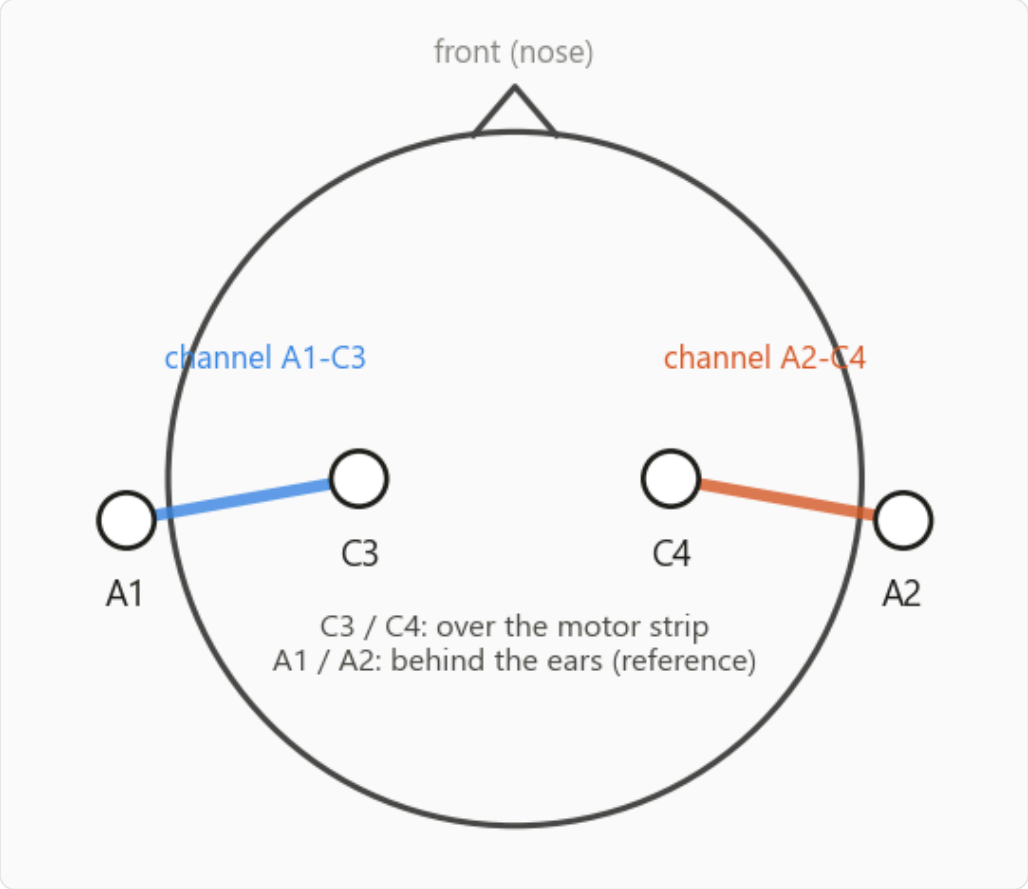


Where each number is computed. Only the green boxes are our own analysis.

There are **two separate analysis paths** in our code, because the SDK's spectrum cannot show the slowest and fastest waves (section 7):

Path	Input	Produces	Compared with
Stress path	SDK spectrum, 0.1 Hz bins, 1–30 Hz	Stress, Fatigue	your first 2 minutes of work
Bands path	raw EEG, 250 samples/s	Brain-wave bars	your eyes-open rest baseline
(vendor)	SDK internal	Attention, Cognitive load, Relaxation	the SDK's own baseline

2. The headset and its signal



Electrode positions seen from above. Each channel is the voltage between an ear electrode and a scalp electrode.

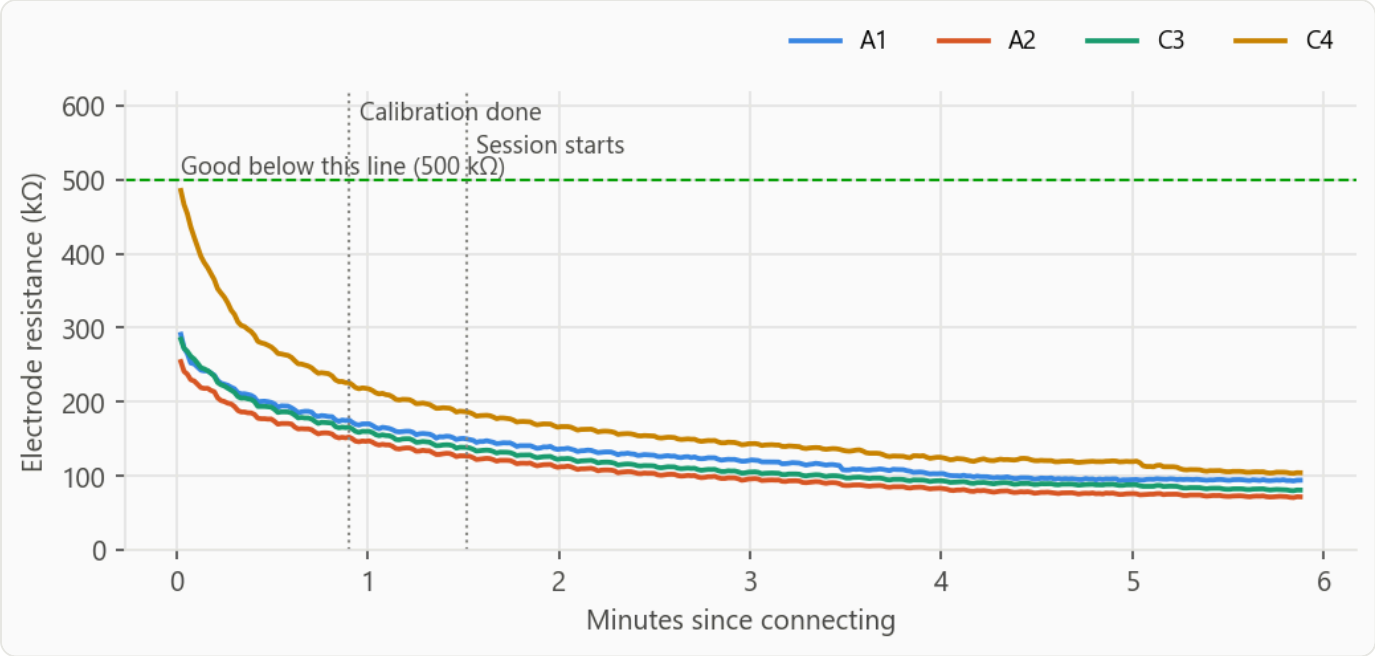
HOW	Four dry electrodes: A1 and A2 behind the ears, C3 and C4 on top of the head. The SDK is set to **bipolar** mode, so it reports two channels: **A1-C3** (left) and **A2-C4** (right). Each channel is sampled **250 times per second**.
WHY	A bipolar channel subtracts two electrodes, which cancels noise they share (for example from the body or the room). The SDK's classifiers expect bipolar mode.
FROM WHERE	Vendor Channel names, sample rate and mode come from the SDK at connection time (`Device.get_channel_names` , `get_eeg_sample_rate`). Electrode names follow the international 10-20 system Literature . Your device: "Headphone", serial 830817.

3. Connecting

HOW	The app asks the SDK's `DeviceLocator` to search for 10 s, lists what it finds, and connects to the one you pick. If connecting fails it retries after 2, 5 and 10 s. Everything the SDK reports (new data, status changes) arrives through *callbacks*, which only run when the app calls the SDK's `update()` ; the app does that every 20 ms from the window's own timer.
WHY	The SDK is single-threaded: calling it from one place avoids crashes and lets the window stay responsive. Bluetooth connections sometimes fail on the first try, so automatic retries save the user a restart.
FROM WHERE	Vendor Connection flow from the SDK documentation (*Capsule flow*, *Device connection* pages) and the vendor's Python examples. Design choice The 10 s scan and the 2/5/10 s retry delays.

4. Electrode contact (the Signal step)

HOW	The SDK measures each electrode's **resistance** (how well it touches the skin). The app grades it: **Good** below 500 kΩ, **Fair** 500–1000 kΩ, **Poor** above. You can continue once every electrode has been Good or Fair for 5 seconds in a row. During the session, the SDK also flags each moment as artifact or clean; if more than 30% of the last 30 s is flagged, the screen says "Adjust headphones" and those seconds are left out of your averages.
WHY	Poor contact lets in noise that looks like brain activity. The 5-second hold stops a single lucky reading from opening the gate.
FROM WHERE	Vendor The 500 kΩ / 1 MΩ thresholds are quoted from the SDK documentation page *Devices modes* (`Docs/html/devices_modes.html`): "Above 1000 kOhm - bad quality, 500 to 1000 kOhm - average quality, Below 500 kOhm - good quality". Artifact flags: SDK `EEGArtifacts` . Design choice The 5 s hold, the 30 s window and the 30% warning level.



Your 13:52 session: dry electrodes keep improving for several minutes after you put the headset on.

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5. Calibration: your alpha peak

HOW	You close your eyes for 30 seconds. The SDK's `CalibrateIndividualNFBQuick` finds the frequency where your alpha rhythm is strongest: the **individual alpha peak frequency (iAPF)**. It returns the peak and an individual alpha range. A result code tells whether it worked (0 = OK, 1 = too many artifacts, 2 = peak on the edge of the range).
WHY	Alpha is strongest with eyes closed, so that is the easiest moment to find its peak. People's alpha peaks differ (roughly 8–13 Hz) and the SDK tunes its later calculations to yours.
FROM WHERE	Vendor Method and the 30-second closed-eyes procedure: SDK documentation page *Calibration* ("CalibrateIndividualNFBQuick will require 1 stage: Closed eyes - 30 sec") and the vendor's *Instruction Manual.docx*. Literature The vendor cites Bazanova (2012) and Posthuma et al. (2001) for iAPF. Your result: **11.7 Hz** (13:52 session).

6. Rest baseline (eyes open)

HOW	You look at a cross for 30–90 seconds. The SDK's Productivity module records your resting levels (`StartBaselineCalibration`) and moves on by itself. At the same time, our code stores the median power of each brain-wave band from the raw EEG; that is the reference for the brain-wave bars.
WHY	All scores are relative. "Attention 70" only means something compared with how you were at rest.
FROM WHERE	Vendor SDK *Productivity metrics* page and Instruction Manual ("Immediately after the 30-second window, click Calibrate Baseline to set the reference point"). Design choice Using the same period as the band reference.

7. The SDK spectrum (PSD)

A **power spectrum** (PSD, power spectral density) splits the signal into frequencies and says how strong each is. The SDK sends one for each channel about 23 times per second, in **0.1 Hz bins** from 1.0 to 30.0 Hz: 291 numbers per channel. One **bin** is one 0.1 Hz slice.

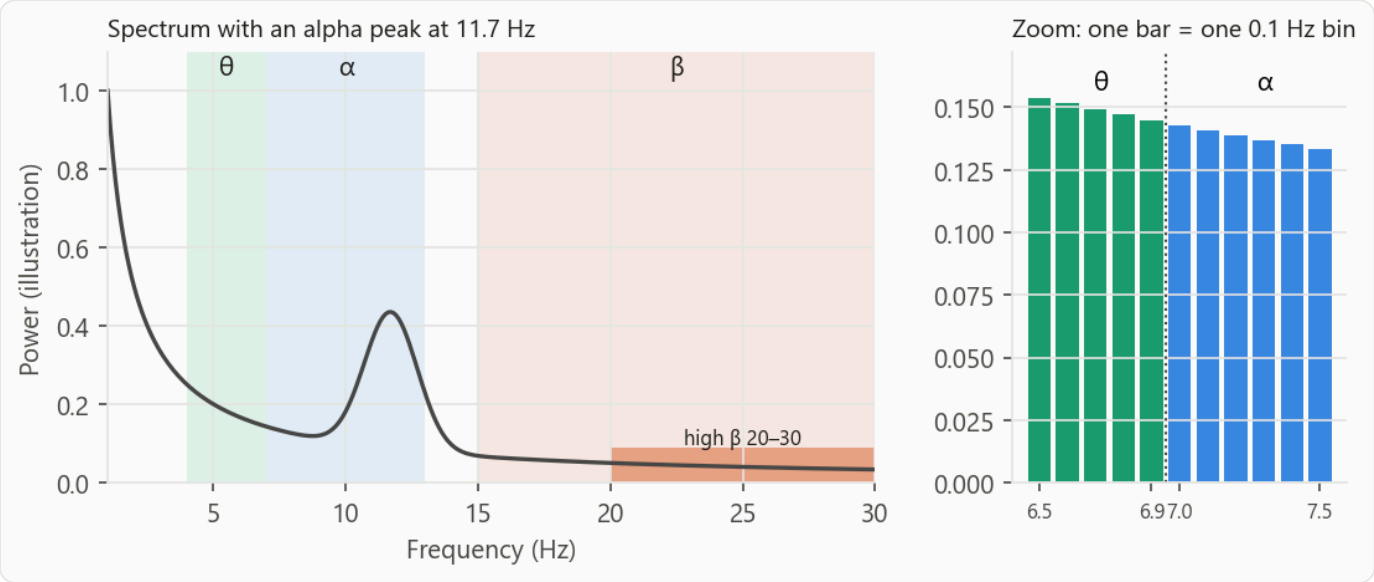
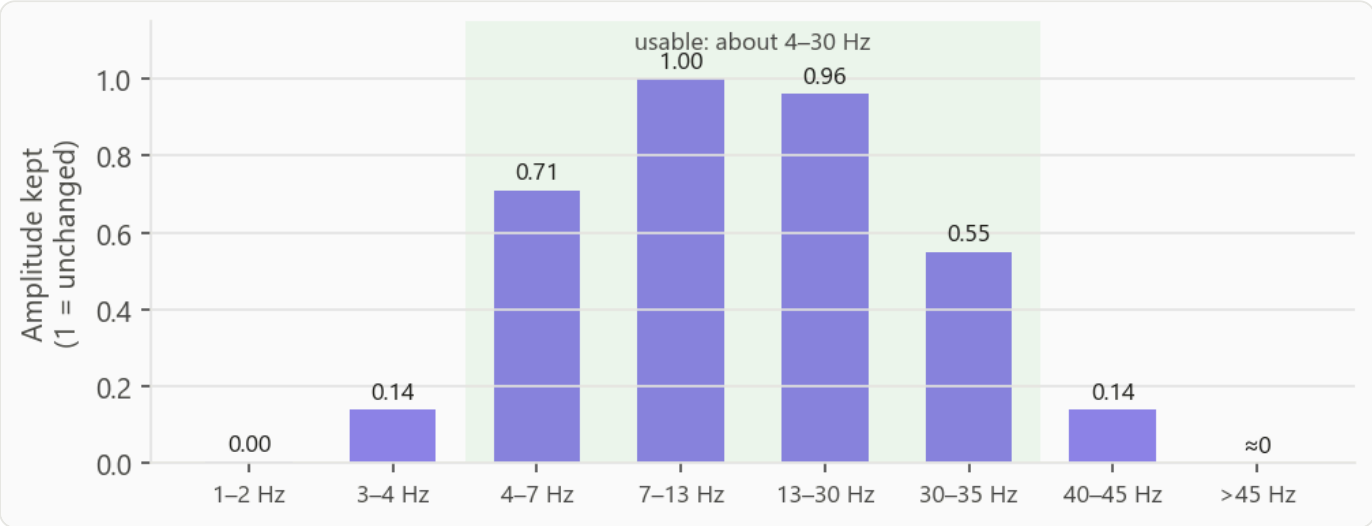


Illustration: a typical EEG spectrum (falling with frequency, with an alpha bump) and the vendor band ranges. Right: the individual bins at the theta/alpha border.

HOW	The app reads 2 spectra per second. For each band it adds the bins with `lo ≤ f < hi` (θ 4.0–6.9: 30 bins; α 7.0–12.9: 60; β 15.0–29.9: 150; high β 20.0–29.9: 100; total 1.0–29.9: 290), divides each band by the channel's total (relative power), and averages the two channels.
WHY	Dividing by the total removes changes in overall loudness (contact getting better or worse), leaving the balance between rhythms. The "< hi" rule stops a border bin from counting twice.
FROM WHERE	Vendor Band ranges from the vendor's *Metrics API 2024.xlsx* (θ 4–7, α 7–12/13, β 15–30 Hz; SMR 13–15 Hz is a separate vendor band) and the SDK's own band edges. We kept them so our ratios match the vendor's formulas. Measured Bin spacing and count read from the SDK. Measured The SDK filters its signal: measured on the simulator (white noise in, compare processed with raw), the chart below. The SDK documentation confirms an "effective EEG measurement bandwidth set from 4 to 35 Hz".



How much of each frequency survives the SDK's filter (measured). Below 4 Hz and above 30 Hz almost nothing is left, so this spectrum cannot show delta or gamma.

8. Attention, Cognitive load, Relaxation (vendor)

HOW	These three come straight from the SDK's **Emotions** classifier as 0–100 values. Our code only clips them to 0–100 and shows the median of the last 5 seconds.
WHY	The vendor designed and tuned these for this headset. Keeping them lets you compare with the vendor's own apps.
FROM WHERE	Vendor SDK *Emotions* page (fields attention, relaxation, cognitiveLoad, cognitiveControl, selfControl). The vendor's *Capsule metrics_definitions.xlsx* describes them as **normalised β/α** (attention), **normalised α/β** (relaxation) and **normalised θ/α** (cognitive load), citing Raufi & Longo (2022) for θ/α Literature . The exact normalisation is not published. Measured Checked on your first session: the vendor's raw concentration and relaxation match our β/α and α/β (Spearman ρ = 0.84 and 0.85), but the 0–100 versions only moderately (ρ ≈ 0.5–0.6).

> **Limit.** In the 18:09 session the vendor's cognitive load rose from 42 to 70 while θ/α of the same spectrum fell from 1.10 to 0.79. We cannot see why inside the DLL.

9. Stress

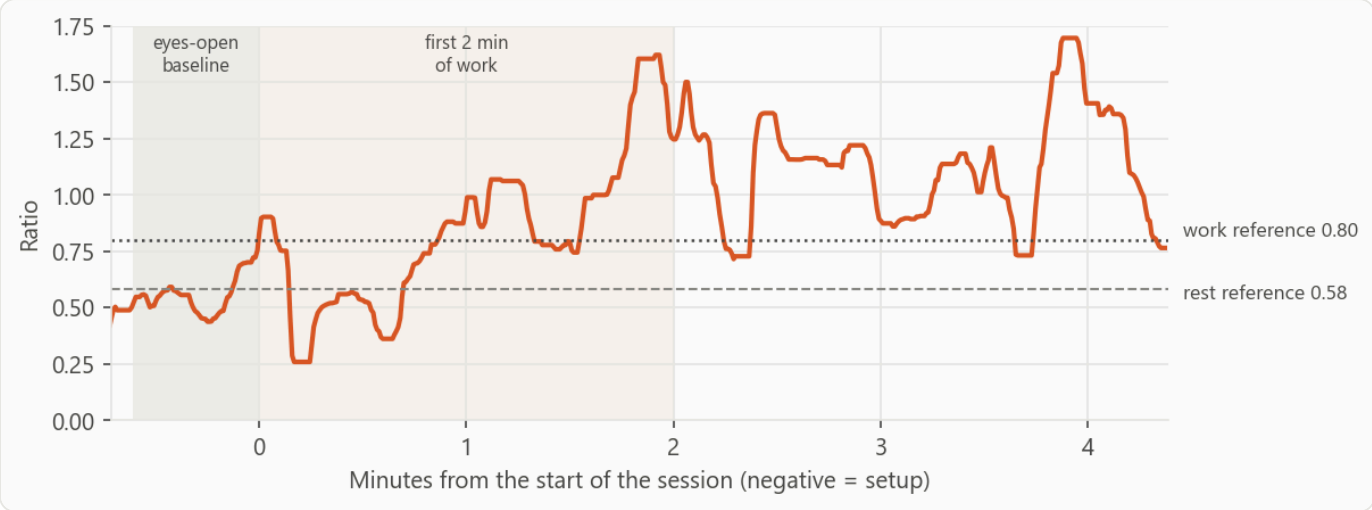
Step 1: the ratio

```
arousal = relative high β (20–30 Hz) / relative α (7–13 Hz)
```

HOW	Computed from each SDK spectrum, 2 times per second.
WHY	Fast beta rises and alpha falls when you are tense or aroused, so their ratio goes up.
FROM WHERE	Literature High-β relative to α is a common arousal/anxiety marker. But a meta-analysis of stress EEG studies (Vanhollebeke et al., 2022) found *no consistent* beta effect and mixed alpha effects, and EEG above ~20 Hz contains muscle activity (Whitham et al., 2007). Design choice Using this ratio as the stress score. Treat it as "arousal compared with your own normal", not as a clinical stress measure.

Step 2: the reference, your first 2 minutes of work

HOW	For the first 120 s of the session (breaks and poor-signal moments do not count) the app stores every ratio and takes the **median**. The Stress card shows "Starts in m:ss" meanwhile.
WHY	Measured In your 13:52 session the ratio was about 0.58 while looking at the cross and about 1.1 during screen work. Compared with rest, ordinary work already looked like stress (the first version showed a stress episode for 3.5 of 4.4 minutes). Looking at a busy screen lowers alpha. The median ignores single spikes.
FROM WHERE	Measured The mismatch was found in your recording. Literature Time-on-task studies also compare with the start of the task rather than with rest. Design choice The 2-minute length.

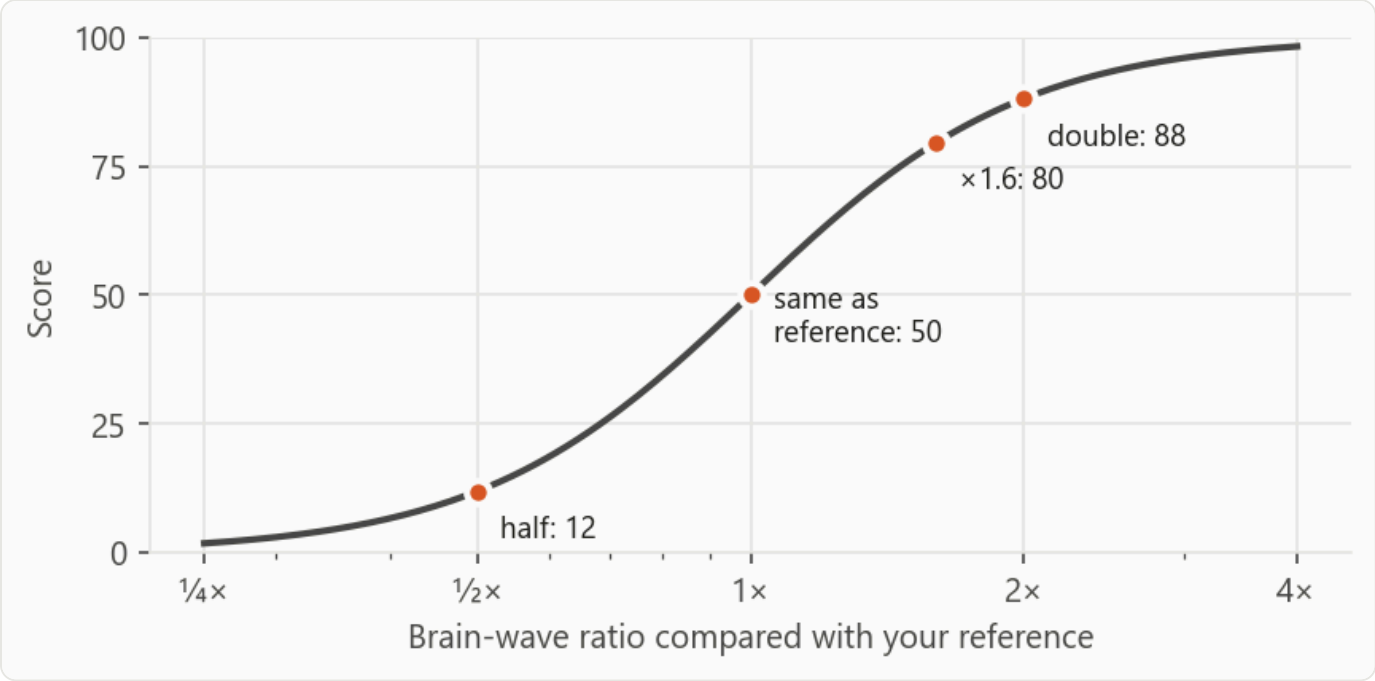


Your 13:52 session: high β / α at rest (grey) was well below its level during work. The work reference sits in between.

Step 3: ratio to score

```
score = 50 + 50 × tanh( log₂( ratio / reference ) )
```

HOW	$\log_2$ turns "double" into +1 and "half" into −1, so rises and falls are symmetric. tanh squeezes any value into −1…+1, so the score stays within 0–100 and one wild reading cannot jump to the end.
WHY	A fixed, simple mapping everyone can check by hand, with 50 always meaning "same as your reference".
FROM WHERE	Design choice Our own mapping. Check points: ½× → 12, 1× → 50, 1.6× → 80, 2× → 88.



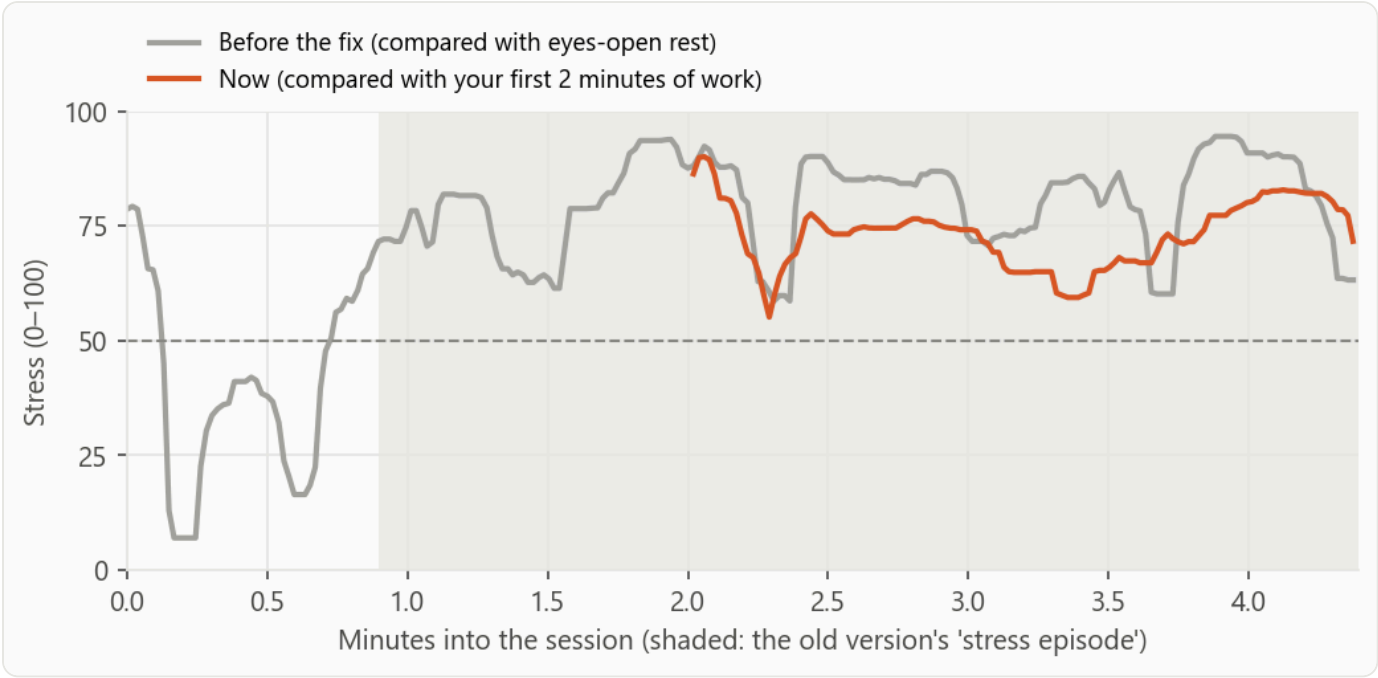
The mapping from ratio to score.

Worked example (13:52 session, 3 minutes in). Relative powers: θ 0.195, α 0.336, β 0.412, high β 0.286.

```
arousal   = 0.286 / 0.336            = 0.850
ratio/ref = 0.850 / 0.798            = 1.066
log₂      = 0.093      tanh = 0.092
score     = 50 + 50 × 0.092           = 54.6
```

Step 4: smoothing and stress episodes

HOW	The card shows the median of the last **30 s** of scores. A **stress episode** starts when that stays ≥ 80 for 30 s and ends when it stays < 65 for 30 s. Episodes end when a break starts.
WHY	Measured On your recording, even 10-second medians of the ratio swing by about ±47% (standard deviation 0.56 in $\log_2$). The first version's thresholds (70 for 15 s) sat inside that normal wobble. Different start/end levels (hysteresis) stop an episode flickering on and off.
FROM WHERE	Measured Chosen by replaying your 13:52 recording under several settings: with these, ordinary work gives no episode. Design choice Not yet validated against a real stress task (plan.md §6.5).



Stress on the 13:52 recording with the first version (grey) and the current one (orange).

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10. Fatigue

```
fatigue ratio = (relative θ + relative α) / relative β
```

HOW	Same steps as stress: SDK spectrum → ratio → reference = median of the first 2 minutes of work → the same 0–100 mapping → median of the last **60 s**.
WHY	Slow waves (θ, α) grow relative to β as you tire during a long task. Tiredness builds over minutes, so a longer smoothing window suits it.
FROM WHERE	Vendor The formula (α+θ)/β is the vendor's Fatigue Score (*Capsule metrics_definitions.xlsx*, citing Tyas et al., 2020, and Jap et al., 2009 Literature). Literature Theta and alpha rise with time on task (Psychophysiology, 2021). Design choice 60 s smoothing and the work reference.

Worked example (same moment):

```
ratio  = (0.195 + 0.336) / 0.412 = 1.288
vs ref = 1.288 / 1.076 = 1.197   →   score = 62.7
```

> **Why not the vendor's fatigue?** The SDK also offers "accumulated fatigue". In your first session it grew at its fastest rate whenever concentration was above 1.7, and your resting concentration was already 2.2, so it reached 6/10 in 3.3 minutes and asked for a break. It is still saved in `productivity.csv` but not used.

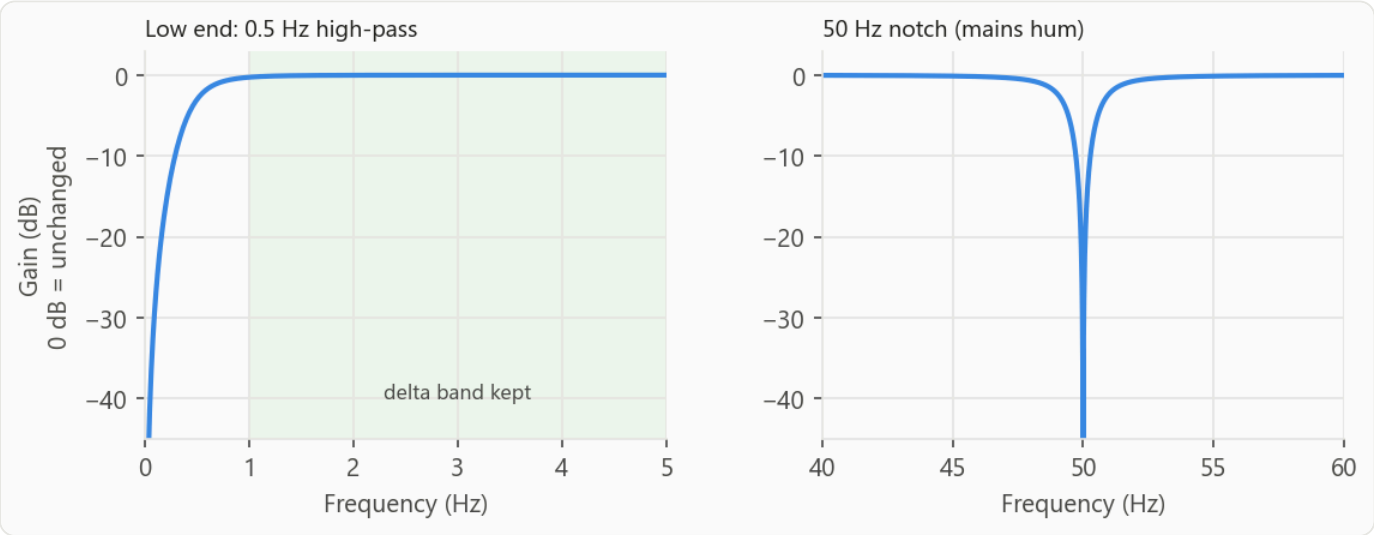
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11. Brain-wave bars (our own spectrum)

The SDK spectrum has no delta or gamma left (section 7), so this path starts from the **raw** EEG, which the SDK passes through unfiltered (1–120 Hz according to its documentation).

Step 1: filter

HOW	A 2nd-order Butterworth **high-pass at 0.5 Hz** plus a **notch at 50 Hz** (Q = 30), combined into one filter and run continuously, block by block, keeping the filter's memory between blocks.
WHY	The high-pass removes very slow drift (sweat, electrodes settling) while keeping delta (1–4 Hz). The notch removes hum from the mains wiring, which in India is 50 Hz.
FROM WHERE	Literature Standard IIR designs from SciPy (`butter` , `iirnotch` , `sosfilt`). Measured Earlier raw recordings from these headphones had 50 Hz hum up to 290× stronger than the 40–45 Hz level. Measured Tested: hum reduced by more than 30 dB and does not leak into gamma.

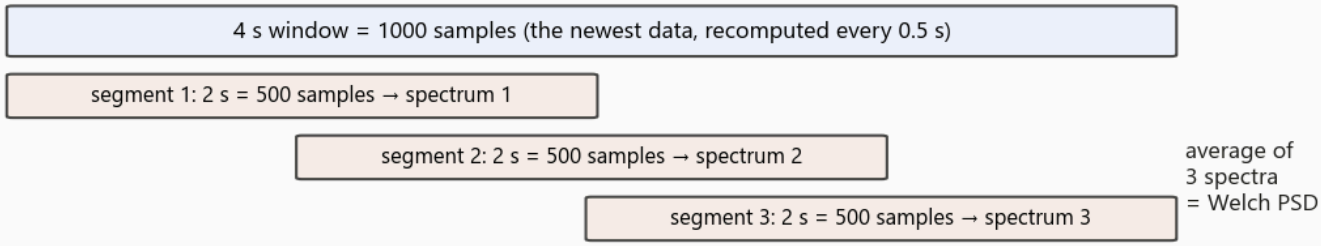


What the filter keeps (0 dB) and removes, computed from the app's own filter.

Step 2: windows and rejection

HOW	Every 0.5 s, take the newest 4 s (1000 samples). Skip the window if the SDK flagged an artifact inside it, or if its peak-to-peak amplitude is more than 4× the median of earlier windows (after the first 8).
WHY	Blinks and movements add huge slow waves that would swamp the real ones. A relative threshold is used because dry electrodes give hundreds of µV even when clean, so a fixed µV limit would reject everything.
FROM WHERE	Vendor SDK artifact flags. Design choice 4 s, 0.5 s step, 4× and 8 windows.

Step 3: Welch spectrum



Welch's method: split the window into overlapping pieces, take a spectrum of each, average them.

HOW	Three 2-second segments with 50% overlap, each tapered with a Hann window and linearly detrended, then averaged. Bin spacing = 1 / 2 s = **0.5 Hz**.
WHY	A single spectrum of noisy data jumps around; averaging three is steadier. 2 s segments are the trade-off between detail (0.5 Hz bins, enough to separate the bands) and steadiness.
FROM WHERE	Literature Welch (1967), implemented by SciPy `signal.welch`. Design choice Segment and window lengths.

Step 4: band power

```
band power (µV²) = Σ PSD(f) × Δf   over lo ≤ f < hi,   Δf = 0.5 Hz
δ 1–4 (6 bins) · θ 4–8 (8) · α 8–13 (10) · β 13–30 (34) · γ 30–45 (30)
```

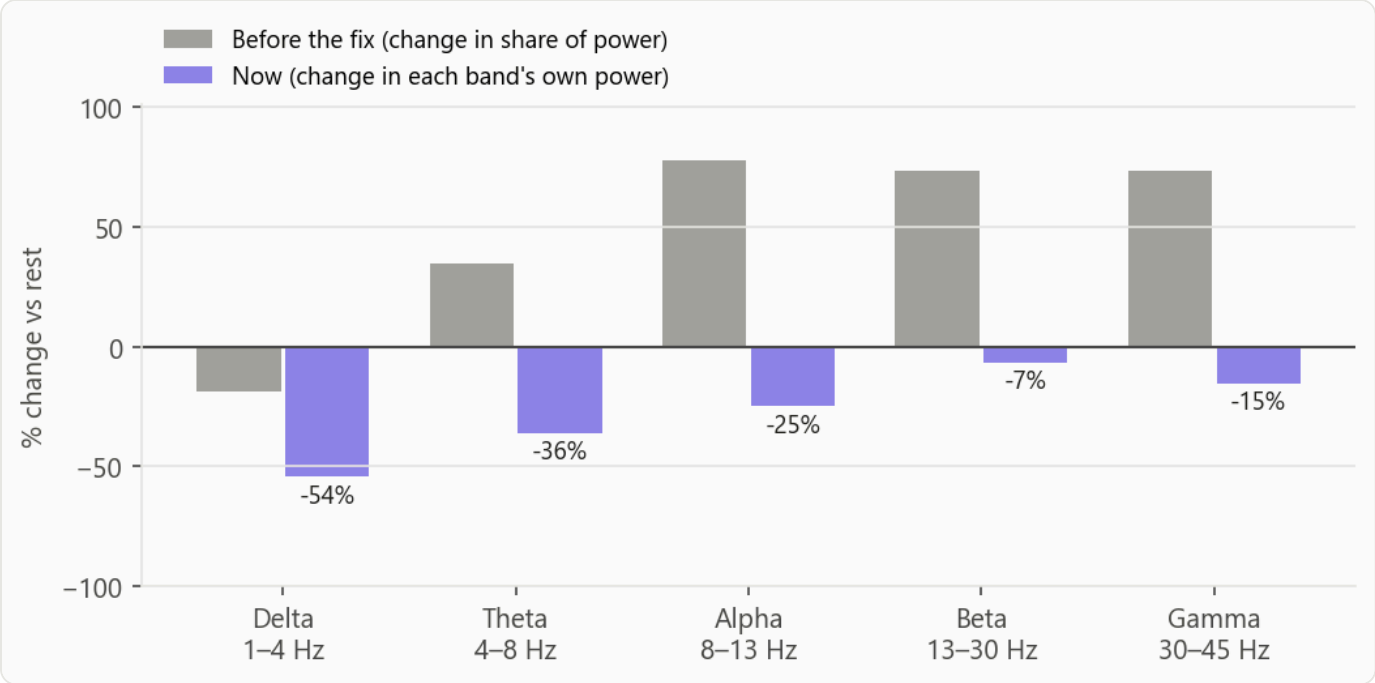
HOW	The PSD is a density (μV^2 per Hz); multiplying by the bin width gives power. The two channels are averaged.
WHY	These bands are the textbook ones, so the chart reads like any EEG textbook. γ stops at 45 Hz, below the 50 Hz hum.
FROM WHERE	Literature Band definitions from the IFCN glossary (Kane et al., 2017). They differ slightly from the vendor ranges in section 7, which are only used for the stress path.

Step 5: compare with rest

```
bar = ( median power in the last 5 s / median power during the rest baseline − 1 ) × 100 %
```

HOW	Each band is compared with its *own* power at rest. +50% = one and a half times, −50% = half.
WHY	Measured The first version compared each band's *share* of total power. During your rest baseline delta held 75% of the power (freshly placed electrodes). As the electrodes settled, delta shrank and every other band's share rose automatically, so alpha, beta and gamma all showed about +75%.
FROM WHERE	Literature Percent change of band power from a reference period is the standard event-related (de)synchronisation measure (Pfurtscheller & Lopes da Silva, 1999). Measured 5 s smoothing: on your data the bars then move about 2 points per second.

	Delta	Theta	Alpha	Beta	Gamma
Rest (μV^2)	229.5	21.7	11.6	26.2	16.2
Session (μV^2)	104.8	13.9	8.7	24.5	13.8
Bar	**-54%**	**-36%**	**-25%**	**-7%**	**-15%**



Your 13:52 session: the first version (grey) against the current one (purple).

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12. Breaks and check-ins

HOW	A break prompt appears when **(A)** 50 minutes pass without a real break, or **(B)** the fatigue score stays ≥ 70 for 3 minutes in a row after the first 10 minutes. At most one prompt per 10 minutes, so an ignored prompt repeats. A break must last 2 minutes to restart the 50-minute timer. Nothing is prompted during a break or for 10 minutes after it, and fatigue measured during a break does not count. Every 10 minutes a sleepiness check-in (1–9) appears, except during a break.
WHY	The timer is a safety net that works even when the EEG cannot see tiredness; the fatigue rule adds a personal signal. The check-in answers are the ground truth for testing whether the fatigue score works.
FROM WHERE	Literature Karolinska Sleepiness Scale (Åkerstedt & Gillberg, 1990). Research on rest breaks and EEG fatigue (Int. J. Psychophysiology 2023; Rest Assured, CHI 2026) motivated regular breaks. Measured The 2-minute minimum: in your session a 1.5-second break had restarted the timer. Design choice 50 min, 70 for 3 min, 10 min cooldown: common-sense defaults, not yet validated.



Illustration (made-up data) of the two break rules.

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13. Recording

HOW	Each session gets a folder named by date and time with CSV files (one row per reading) and a `session.json` summary: `scores.csv` at 1 per second, `bands.csv` at 2 per second, `events.csv` for markers, breaks, prompts and episodes, plus the SDK's raw outputs.
WHY	Everything on screen can be recomputed and checked later. This explainer's figures are rebuilt from these files.
FROM WHERE	Design choice File layout. Python's `csv` and `json` modules.

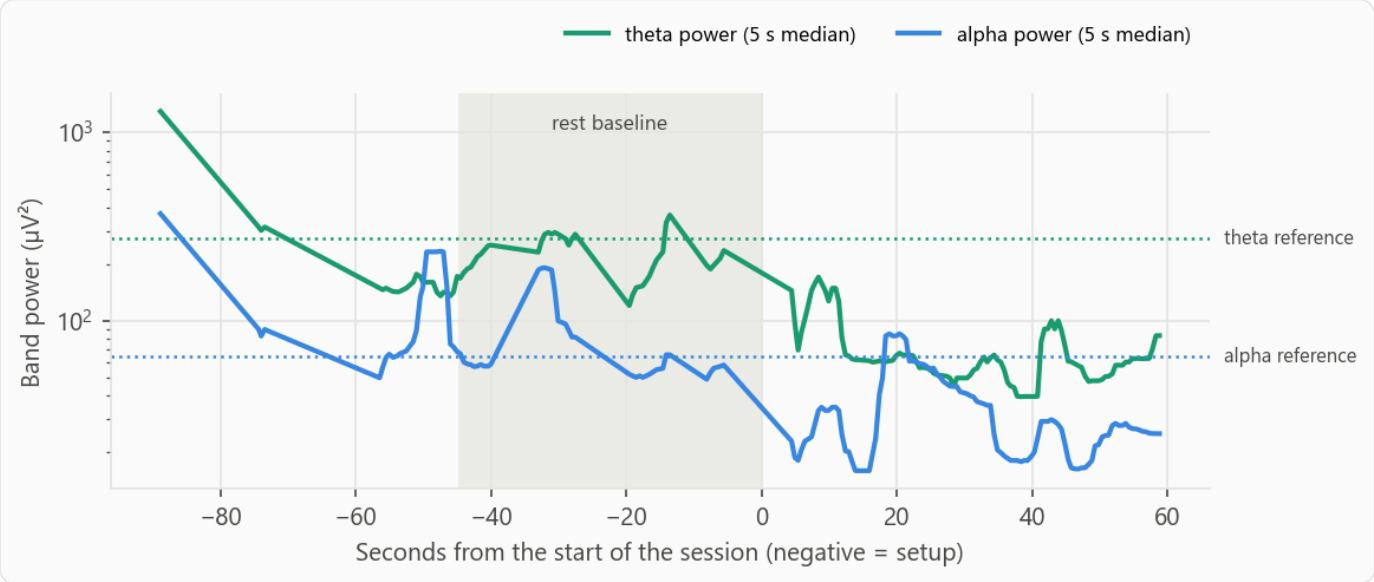
14. Testing

HOW	80 automated tests (pytest). Unit tests feed made-up signals with known answers; simulator tests run the real SDK with its simulated devices; a self-test drives the real window through all seven steps.
WHY	A known input must give a known output before real brain data is trusted.
FROM WHERE	Measured Examples: white noise must give each band a share equal to its width (within 3%); a pure 10 Hz tone must be ≥ 95% alpha; the notch must cut 50 Hz by more than 30 dB; slow drift at rest must not move the other bands. Vendor The SDK's *Noise* and *SinWave* simulated devices.

15. Known issues

A noisy rest baseline makes every bar negative

In the 18:09 session the SDK flagged 49 of 90 windows during the eyes-open baseline as artifacts, and the slow waves were several times stronger than a minute later (theta 272 μV^2 at rest against 60 μV^2 in the session). The rest reference was therefore far too high, so during the session every band read lower (delta −92%, theta −83%, alpha −69%). The amplitude check did not catch it: it learns "normal" from the first windows, and the signal was already large then.



18:09 session: theta and alpha power over time. The dotted lines are the references frozen from the rest baseline (grey); the session sits far below them.

Proposed fix (not yet built): check the baseline's quality before accepting it, and repeat it if more than about 20% is artifact.

Vendor scores are a black box

Attention, cognitive load and relaxation cannot be checked step by step (section 8). A possible change is to replace them with our own β/α and θ/α computed like stress.

Stress and fatigue are not yet validated

The thresholds were tuned on short test sessions. The planned check (plan.md §6.5) is mental arithmetic vs rest for stress, and the sleepiness check-ins vs the fatigue score over long sessions.

16. All numbers at a glance

Setting	Value	From where
Channels, sample rate	A1-C3, A2-C4 bipolar, 250 Hz	Vendor
Contact grades	Good < 500 kΩ, Fair ≤ 1 MΩ, held 5 s	Vendor thresholds, Design choice 5 s
Calibration	30 s eyes closed, Quick iAPF	Vendor
SDK spectrum	0.1 Hz bins, 1–30 Hz, read 2×/s	Vendor Measured
Vendor bands (stress path)	θ 4–7, α 7–13, β 15–30, high β 20–30 Hz	Vendor
Stress ratio	high β / α	Literature Design choice
Fatigue ratio	(θ + α) / β	Vendor Literature
Reference	median of first 120 s of work	Measured Design choice
Score mapping	$50 + 50 \cdot \tanh(\log_2(\text{ratio}/\text{ref}))$	Design choice
Smoothing	vendor scores 5 s, stress 30 s, fatigue 60 s, bars 5 s	Measured Design choice
Stress episode	≥ 80 for 30 s, ends < 65 for 30 s	Measured Design choice
Level words	Low < 35 ≤ Moderate < 65 ≤ High	Design choice
Raw filter	0.5 Hz high-pass (order 2), 50 Hz notch (Q 30)	Literature Measured
Welch	4 s window, 2 s Hann segments, 50% overlap, every 0.5 s	Literature Design choice
Rejection	SDK artifact, or peak-to-peak > 4× median	Vendor Design choice
Textbook bands (bars)	δ 1–4, θ 4–8, α 8–13, β 13–30, γ 30–45 Hz	Literature

Break prompt	50 min, or fatigue ≥ 70 for 3 min after 10 min; 10 min cooldown; break ≥ 2 min	Design choice Measured
Check-in	KSS 1–9 every 10 min	Literature

17. Check yourself

Try to answer from memory before looking at the answers below. If one is hard, reread the section in brackets.

1. Why can't the brain-wave bars use the SDK's spectrum? (7, 11)
2. The SDK spectrum has 0.1 Hz bins. How many bins does the alpha band (7–13 Hz) add up, and why is the 13.0 Hz bin not one of them? (7)
3. A ratio is 1.5× your reference. Without a calculator, is the score above or below 80? (9)
4. Why is stress compared with your first 2 minutes of work and not with the eyes-open rest? (9)
5. Our Welch segments are 2 s long. What bin spacing does that give, and what would 4 s segments give? (11)
6. Delta drops by half while the other bands stay the same. What would the old share-based chart have shown for alpha, roughly? What does the current chart show? (11)
7. You take a 90-second break at minute 45. When does the 50-minute prompt come? (12)
8. Which three numbers on screen can you *not* recompute by hand from the saved data, and why? (8)

Answers

1. The SDK filters its signal to about 4–30 Hz before computing the spectrum, so almost no delta (gain 0.004 at 1–2 Hz) or gamma is left.
2. 60 bins (7.0, 7.1 … 12.9). The rule is lo ≤ f < hi, so 13.0 is outside; that keeps a border bin from counting in two bands.
3. Below. 1.6× gives 80, so 1.5× gives a little less: $\log_2 1.5 = 0.585$, tanh = 0.526, score ≈ 76.
4. Looking at a busy screen lowers alpha, so high β / α is higher during any screen work than at rest. Compared with rest, ordinary work already looked like stress (first session: an "episode" for 3.5 of 4.4 minutes).
5. 1 / 2 s = 0.5 Hz. 4 s segments would give 0.25 Hz bins, but only one segment would fit in the 4 s window, so the spectrum would be much noisier.
6. If delta was most of the power (say 75%), halving it makes the total smaller, so every other band's share rises by roughly half (about +60%). The current chart compares each band with its own rest power: alpha 0%.
7. At minute 50. 90 s is shorter than the 2-minute minimum, so the timer was not restarted. (A 2-minute break would have moved it to about minute 97.)
8. Attention, Cognitive load and Relaxation. They come from the vendor's closed Emotions classifier, whose normalisation is not published.

18. Sources

Vendor (in `D:\trainsvm\eeg_headphones\Capsule_2.0.59`)

- Capsule API documentation, `CapsuleAPI/Docs/html`: *Devices modes* (resistance thresholds, bipolar mode), *Calibration* (Quick iAPF), *Productivity metrics*, *Emotions*, *PSD*, *Raw and raw filtered signal* (1–120 Hz raw, 4–35 Hz filtered).
- *Capsule metrics_definitions.xlsx* and *Metrics API 2024.xlsx*: metric formulas, band ranges and the studies the vendor cites.
- *Instruction Manual.docx*: calibration and baseline procedure.

Published work

- Kane N. et al. (2017). A revised glossary of terms most commonly used by clinical electroencephalographers. IFCN. *Clin. Neurophysiol. Practice*.
- Vanhollebeke G. et al. (2022). The neural correlates of psychosocial stress: a systematic review and meta-analysis of spectral analysis EEG studies. *Neurobiology of Stress*.
- Whitham E.M. et al. (2007). Scalp electrical recording during paralysis: EEG frequencies above 20 Hz are contaminated by EMG. *Clin. Neurophysiol.*
- Raufi B., Longo L. (2022). An evaluation of the EEG alpha-to-theta and theta-to-alpha band ratios as indexes of mental workload. *Front. Neuroinform.*
- Pfurtscheller G., Lopes da Silva F.H. (1999). Event-related EEG/MEG synchronization and desynchronization: basic principles. *Clin. Neurophysiol.*
- Welch P. (1967). The use of the fast Fourier transform for the estimation of power spectra. *IEEE Trans. Audio Electroacoustics*.
- Åkerstedt T., Gillberg M. (1990). Subjective and objective sleepiness in the active individual. *Int. J. Neuroscience*.
- Time-on-task EEG: *Psychophysiology* (2021), doi:10.1111/psyp.13805.
- Rest breaks: *Int. J. Psychophysiology* (2023), S0167876023000612; Schick L. et al. (2026) Rest Assured, CHI '26.
- Vendor-cited: Tyas et al. (2020), Jap et al. (2009) for fatigue; Bazanova (2012), Posthuma et al. (2001) for iAPF.

Software

Capsule SDK 2.0.40 (via ctypes), NumPy, SciPy (`butter`, `iirnotch`, `tf2sos`, `sosfilt`, `welch`), Matplotlib, customtkinter, pytest. Code: `cogscore/metrics.py` (stress path), `cogscore/spectral.py` (bands path), `cogscore/rules.py` (breaks, episodes), `cogscore/config.py` (every number above).